

Emerging contaminants (ECs), including per- and polyfluoroalkyl substances (PFAS), microplastics, pharmaceuticals and personal care products (PPCPs), and pesticides, pose escalating threats to environmental quality and human health in drinking water, wastewater, and contaminated soils. Their extreme persistence, complex multiphase behavior, and resistance to conventional chemical treatment create urgent challenges for ensuring safe drinking water and restoring contaminated soils. My research addresses these challenges by advancing physical–chemical treatment technologies that operate without chemical additives, enabling cleaner reaction pathways, minimizing byproduct formation, and penetrating inhibitory matrices where chemical oxidants often fail. Through the integration of non-thermal plasma (NTP) systems, architected adsorbent materials, and circular regeneration strategies, my work seeks to transform how persistent contaminants are destroyed across the environment, laying the foundation for sustainable environmental remediation.

Previous and Current Research

My doctoral research established my foundation in aquatic chemistry and advanced treatment. I synthesized bio-coagulant from *Opuntia Ficus-Indica*, a cactus species, for pretreatment of oil sands process-affected water (OSPW) at the University of Western Ontario, Canada.¹ Additionally, I employed its biomass to synthesize activated biochar, which was then used for textile wastewater treatment.² I also designed and operated a non-thermal pulsed electrical-discharge plasma reactor to degrade reactive textile dyes and endocrine disruptors, including 17 α -ethinylestradiol and the pesticides alachlor and chlorpyrifos. This work combined radical chemistry, degradation kinetics, and byproduct identification to understand how NTP drives degradation pathways in complex waters.³ During my postdoctoral work at Temple University, I designed a continuous-mode NTP reactor with a rotating spark gap for high-voltage pulsing, enhancing contaminant degradation. I also evaluated and enhanced the existing analytical methods for the analysis of active pharmaceutical ingredients (APIs) by assessing the impact of various storage and shipping conditions. Additionally, I managed a USAID project, recruiting and training individuals (faculty and students) from Egypt to establish a Center of Excellence for water resiliency in Egyptian universities in partnership with five other US universities. At UMaine, my research involves developing graphene-based adsorbents with edge functionalization for targeted PFAS removal.⁴ I investigate how oxidation affects the microwave reactivity of graphene for regeneration of PFAS-laden graphene, enabling sustainable reuse. Currently, I am leading projects as a co-PI to develop NTP systems to degrade PFAS from contaminated soils, to regenerate PFAS-laden granular activated carbon, and to degrade gaseous PFAS by-products formed during adsorbent regeneration. I also work with an interdisciplinary team to develop virtual reality tools that simulate contaminant transport in soils for research and educational purposes.

Future research at the EC² Lab at Kansas State University

I plan to establish the **EC² Lab (Environment, Climate, and Circularity)** to develop **chemical-free** sustainable treatment technologies that link energy efficiency, environmental protection, and resource recovery. I will integrate NTP systems with micro and nano-bubble (M&NB) technology for potable **water reuse** and **adsorbent regeneration**, alongside architected adsorbent materials to **immobilize and remove contaminants** in complex water and soil matrices. Together, these approaches establish a circular framework for environmental protection and resource recovery. Outlined below are three primary research thrusts of the EC² Lab, each strategically developed to address a distinct challenge.

Thrust 1: Non-thermal Plasma – Micro/Nanobubble Reactors for Wastewater Reclamation

Motivation: NTP, a destructive technology, holds great promise for the complete mineralization of ECs. NTP chemistry is tunable by adjusting the inlet gases; various oxidative and reductive reactive species can be generated to target a broad range of contaminants. A major challenge is achieving efficient transfer of these reactive species from the gaseous phase to the liquid phase.⁵ Integrating M/NB generators with NTP systems will enhance mass transfer and increase in-situ generation of reactive species in the gaseous phase of water bubbles.^{6,7}

Proposed Research: The project will include fundamental studies of plasma formation, reactive species generation, and diffusion mechanisms at the bubble-liquid interface. A custom experimental setup will include optical emission spectroscopy (OES) to analyze reactive species production and fast-frame imaging camera (picosecond high-speed intensified charge-coupled device (ICCD)) to analyze plasma propagation and stability in bubbles of varying sizes and gas compositions. Additionally, the project will investigate the influence of bubble size and gas type (e.g., air, O₂, H₂ and inert gases) on plasma-induced chemistry at the interface. UPLC/MS/MS, toxicity bioassays, and qPCR will be employed to comprehensively evaluate the treatment process. Additionally, I will apply this system within a **One Water framework** by targeting micropollutants such as PFAS, PPCPs, and antimicrobial resistance (AMR) to generate direct potable reuse (DPR) and indirect potable reuse (IPR) water for irrigation, meeting the growing demand for advanced treatment.

Thrust 2: 3D-Printed Architectural Adsorbent Materials

Motivation: Granular activated carbon (GAC) is a widely used and proven method for removing emerging contaminants from water. However, its effectiveness significantly reduces in the presence of natural organic matter.⁸ Graphene oxide (GO) offers a promising alternative due to its high surface area and sheet-like structure, potentially overcoming these limitations. Nonetheless, the nanoscale size of GO, like other nanoparticles, poses significant challenges for real-world applications due to difficulties in material recovery post-treatment.⁹ Fabrication of 3D printed architectures of nanomaterials via additive manufacturing could bridge this gap, enabling efficient pollutant removal in practical settings.^{10,11} This need extends to soil systems, where high-surface-area recoverable adsorbents are required for in situ remediation.

Proposed Research: I will fabricate 3D-structured porous graphene-based architected adsorbent utilizing the direct ink writing (DIW) additive manufacturing technique. GO powder will be mixed with cellulose nanocrystals (CNCs) and amine-terminated polylactide (PLA-NH₂) to produce printing ink with non-Newtonian fluid properties. PLA-NH₂ is chosen for its hydrophobic backbone and amine groups, which introduce positive charges for electrostatic interaction with anionic contaminants.¹² I will also create 3D-printed structures for in situ soil remediation to immobilize PFAS in contaminated farms. These larger architected designs will be recoverable after being deployed in the soil, allowing for regeneration and reuse. This offers a significant advantage over current biochar applications, which do not allow for such recovery.

Thrust 3: Circular Regeneration of PFAS-Laden Adsorbents

Motivation: Thus far, solid-phase PFAS removal remains highly underexplored. Most aqueous PFAS removal technologies, such as UV or chemical catalysis, are effective only in bulk water and not in the solid phase. Currently, thermal treatment technologies are the dominant methods for PFAS-laden solid waste regeneration.¹³ However, these methods are energy-intensive, centralized, and emit harmful gases (e.g., HF, hydrofluorocarbons, gaseous PFAS). Thus, there is a pressing need for decentralized and emission-safe approaches for regenerating PFAS adsorbents.

Proposed Research: Building on my work with graphene-based adsorbents, PFAS-laden GAC, and plasma-based treatment, in this thrust, I will develop regeneration strategies that act directly on PFAS-laden solid media using NTP and microwave-based processes. I will study how these treatments control PFAS release, breakdown, and mineralization on GAC and graphene-based adsorbents, and I will monitor off-gases and inorganic fluorine to confirm destruction. I will also measure how pore structure, surface chemistry, and adsorption capacity change over repeated regeneration cycles. Finally, I will combine these results with simple life-cycle and cost analyses to identify regeneration approaches that can realistically be adopted by utilities and industry.

Research Funding

I have been actively involved in proposal writing and review, currently serving as an NSF ad hoc reviewer for the ECosystem for Leading Innovation in Plasma Science and Engineering (ECLIPSE) program. Building on this

experience, I plan to pursue the NSF CAREER Award focused on the fundamental science of non-thermal plasma in water treatment, along with proposals through the NSF CBET-1440 and CHE 6882. I will also seek opportunities through the DoD's SERDP and ESTCP initiatives, USDA's NIFA AFRI and ARS programs, and foundations, including Water Research Foundation (WRF), and Bill Melinda Gates Foundation. At the state level, I will pursue support from the Kansas Department of Health and Environment (KDHE) (Kansas Brownfields Program and the Kansas State Revolving Fund for drinking water and wastewater infrastructure), the Kansas Water Office (State Water Plan Fund programs, including the HB 2302 Technical Assistance Grant Fund and Water Projects Grant Fund), the Kansas Department of Agriculture's conservation cost-share programs, Kansas NSF EPSCoR seed funding opportunities, and the Kansas Water Institute/Kansas Water Resources Institute competitive research support through the USGS Water Resources Research Act program.

Resources and Collaboration at Kansas State University

The UNH provides the analytical and fabrication infrastructure needed to advance my research on non-thermal plasma, micro/nanobubble systems, and 3D-printed graphene-based adsorbents. Across all thrusts, I will use core facilities for high-resolution contaminant and byproduct tracking, including UPLC, ICP-MS, GC-MS/MS, and ion chromatography through the Analytical Chemistry Core, and Q Exactive Orbitrap capability for accurate-mass screening. The Chemistry and Biochemistry Central Instrumentation Core Facility provides LC-MS, IR, and X-ray tools to support materials characterization and mechanistic studies. K-State's Libraries Innovation Lab Fabspace (filament and resin 3D printers and laser cutters) and the Technology Development Institute (CNC, waterjet, and prototyping) will enable rapid iteration of architected sorbents and custom reactor components. At K-State, I am excited to collaborate with the Kansas Water Institute cohort and Helwig BAE faculty to connect treatment performance with watershed-scale transport, agricultural interfaces, and circular nutrient recovery, and with the Center for Hazardous Substance Research to engage stakeholders and translate results into technically sound solutions. Together, these resources support launching the EC² Lab and sustaining a multi-thrust research program.

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